CONFIDENTIAL

UTM UNIVERSITI TEKNOLOGI MALAYSIA
RESEARCH UNIVERSITY
School of Computing,
Faculty of Engineering

FINAL EXAMINATION
SEMESTER I, 2020/2021

(PART B)

SUBJECT CODE	**:**	**SECR1213**
SUBJECT	**:**	**NETWORK COMMUNICATIONS**
YEARS / COURSE	**:**	**SECI/J/R/V/D/B**
TIME	**:**	**1 HOURS 30 MINUTES**
DATE	**:**	**16 FEBRUARY 2021**
PLACE	**:**	**ONLINE e-LEARNING**

INSTRUCTIONS :

1. Answer ALL questions.
2. Use the softcopy of answer booklet (.docx) provided for the answers.
3. Candidates may insert answer in jpeg, png formatted for some questions.
4. Save the answer booklet as a pdf formatted and submit to the link in the elearning.

(Please fill in your particulars on the front page of the answer booklet provided)

Name	
I/C No.	
Group / Section	
Lecturer's Name	

This questions paper consists of **4** (**FOUR**) pages including this page.

PART B

STRUCTURED QUESTIONS (40 MARKS)

(INSTRUCTION: Please answer all questions in the answer booklet.)

QUESTION 1 [15 Marks]

(a) Answer the following questions.

i) You are given the IP address 192.168.4.50/28. List ALL the addresses within the same subnet of the given address. [2 *Marks*]

ii) If the IP address 172.16.14.0/24 is divided into 8 equal sized subnet, what is range of usable addresses for Subnet 2 (the second subnet) and their subnet mask? Show your workings. [3 *Marks*]

(b) Consider the following Figure 1 with a connection of 5 routers with the link costs, respectively.

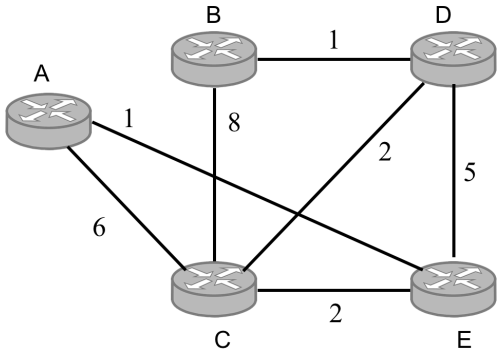


Figure 1

i) Write the complete Bellman-Ford equation for the Router E to Router B [4 *marks*].

ii) Construct the least cost paths for Router E. [2 *marks*]

(c) A datagram with 8000 bytes arrived at a router and must be forwarded to a link with an MTU of 2100. The identification number of the original datagram is 714. Draw all IP fragments that can be generated after fragmentation. [4 *Marks*]

QUESTION 2 [15 Marks]

(a) i) Explain the purpose of backoff algorithm in CSMA/CD. [2 *marks*]

ii) Describe one difference between channel partitioning and random access of multiple access protocols. [3 *marks*]

(b) With regards to ARP table, answer the following questions:

i) Refer to Figure 2 below. If Laptop-A has had some communications with all hosts EXCEPT Laptop-B, what will the ARP table of Laptop-A looks like? Draw the ARP table for Laptop-A (you can ignore the TTL). [2 *Marks*]

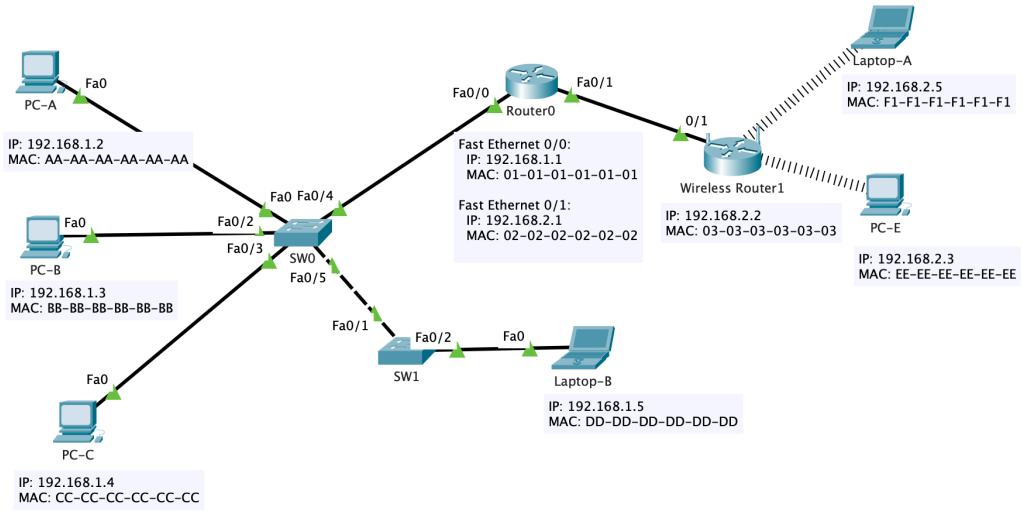


Figure 2

ii) Explain your answer in (i). [3 *Marks*]

(c) A data of 100100 will be sent to the receiver through a network link. Considering the generator of 1101, generate the data that will be sent by the sending host. Show your workings. [5 *Marks*]

QUESTION 3 [10 Marks]

(a) The 802.11 standards require a parameter called beacon frames.

i) Who send this beacon frames and to whom? [1 *Mark*]

ii) What are the TWO most important information contained in the beacon frames? [2 *Marks*]

iii) What are the TWO ways for a wireless device to get connected to an Access Point? [2 *Marks*]

(b) Suppose host A is moved to the visited network from the home network. Show the major steps performed if the mobility is done with the help of indirect routing by drawing, numbering and labelling the arrows to indicate the sequence of the steps. [5 *marks*]

------- End of Question -------

GOOD LUCK !